Wang et al. 2019 — Glioblastoma (Brain)

GEOMETRY: LINEAR (non-branching)

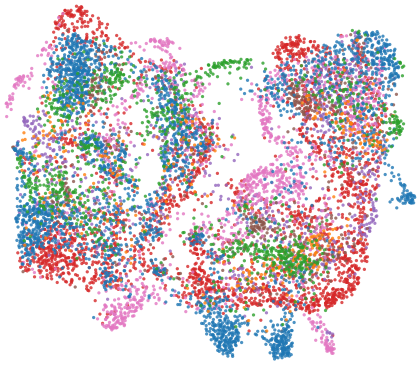
Malignant cell-state manifold · embryonic-program re-analysis (CCCA meta-atlas)

10x scRNA-seq · 9,632 malignant cells · 7 samples (of 10) | raw module means: Gast −0.003 · pSom +0.008 · Crest −0.007

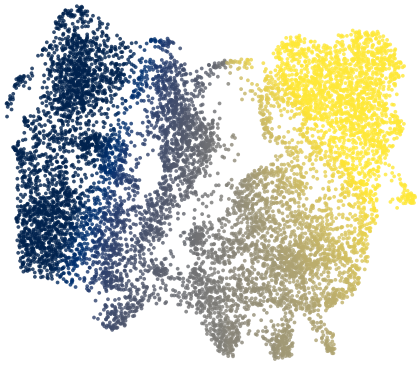
GO / NO-GO GATES

Gate 1	Mixing + continuity	GO	7 samples share ONE continuous manifold (single kNN component after island removal); per-neighbourhood iLISI rises 1.0→2.3/7 across the θ-sweep; Neftel meta-program lobes intermix, not per-sample islands; cycling a scattered ~19% minority.
Gate 2	Module geography	GO	Single collapsed embryonic axis: pSom-high/Crest-low origin → Crest-high terminus (Crest decile mean pt 0.84). Gast/pSom co-vary (+0.16), both oppose Crest (−0.87/−0.49). Non-branching case — GO with Gate 3 = NA.
Gate 3	Hub cross-repression	NA	No embryonic-module bifurcation: PPT fork count is resolution-dependent, not stable — harmony-space 1,3,2,1 (Nodes 20/30/50/100), UMAP-space 1,2,2,2,3 (Nodes 20/30/50/80/120); no tip is Gast-positive (tips are pSom-vs-Crest poles). Node-cluster Gast↔pSom r=+0.07 sits inside the label-shuffle null [−0.35,+0.28] (also inside null on UMAP trees). Neftel AC/OPC/NPC/MES multipolarity ≠ module bifurcation — no branch to test.

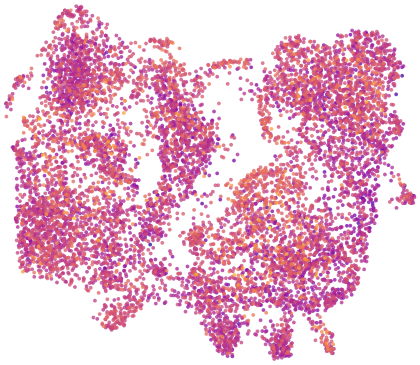
Samples (integration)



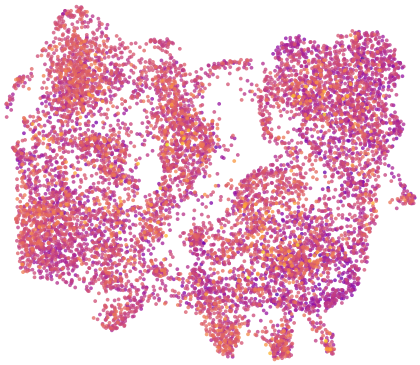
Pseudotime



Gastrulation (raw)



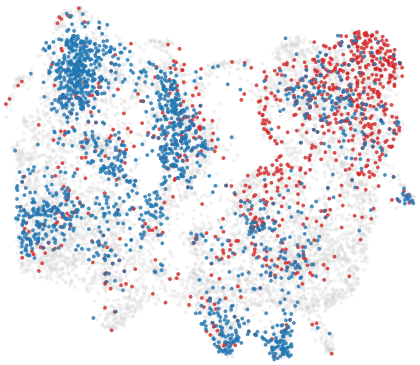
Somitogenesis (raw)



Neural Crest (raw)



EMT-TF: ZEB1 (early) · PRRX1 (late)



PARAMETER CHOICES & RATIONALE

Sample threshold:	≥237 malignant (largest multiplicative count-gap 396→143); 7/10 samples, 9,632 cells
Embedding:	n_pcs=20 (Brain-10x precedent; PC1≈PC2≈11% → 2D cloud), Harmony θ=10 on sample, spread=1.3
Cell cycle:	scored de novo (Tirosh S/G2M — absent in source metadata); orthogonal to modules (|r|≤0.12)
Trajectory:	scFates curve (non-branching), Nodes=20 μ=100 — fold-free (self-approach 2.82, DC1 corr +0.98)
Root:	Gast/pSom-high, Crest-low tip (developmental origin; Crest is late/terminal)
EMT-TFs:	ZEB1 (70%) & PRRX1 (40%) — VIM excluded (marker, not TF); Snail/Twist/FoxC undetectable

VERDICT: The embryonic pattern reproduces as a single collapsed axis — a pSom/Gast-high origin grading to a Crest-high terminus — NOT an embryonic bifurcation. Neftel GBM state-multipolarity does not carry a Gast↔pSom branch that survives a fold-stability + independence null.